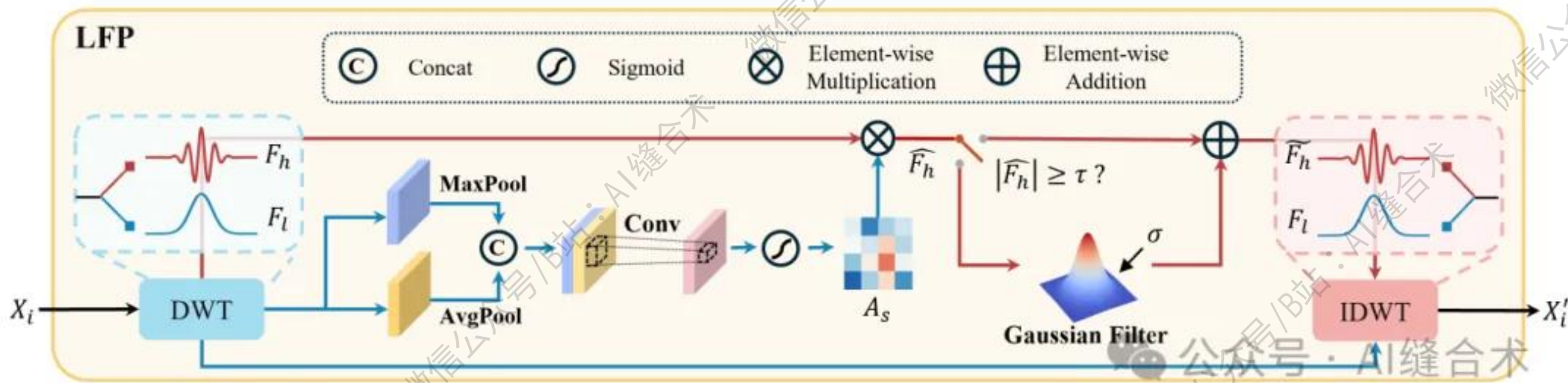


核心速览



论文题目: Seeing Through the Noise: Improving Infrared Small Target Detection and Segmentation from Noise Suppression Perspective

中文题目: 透过噪声：从噪声抑制视角提升红外小目标检测与分割性能

论文链接: <https://arxiv.org/pdf/2508.06878v2>

所属单位: 1 北京航空航天大学 2 南开大学

核心速览:

本文从噪声抑制角度出发,提出一种新型噪声抑制特征金字塔网络(NS-FPN),通过集成低频引导特征净化(LFP, 实际上是在频域进行空间注意力和门控滤波操作)模块和螺旋感知特征采样(SFS)模块,以提升红外小目标检测与分割(IRSTDs)性能,减少误报。

<https://mp.weixin.qq.com/s/ahT0QSIpsDWsxlrPxymVsQ>

```
LFP(  
  (dwt): ConvDWT(  
    (dwt_forward): DWTForward()  
  )  
  (idwt): ConvIDWT(  
    (dwt_inverse): DWTInverse()  
  )  
  (attention): SpatialAttention(  
    (conv1): Conv2d(2, 1, kernel_size=(7, 7), stride=(1, 1), padding=(3, 3), bias=False)  
    (sigmoid): Sigmoid()  
  )  
  (gaussian_filter): LearnableGaussianFilterBank(  
    (sigmas): ParameterList( 0): Parameter containing: [torch.float32 of size 1 (cuda:0)])  
  )  
)
```

```
input_tensor_shape : torch.Size([1, 32, 256, 256])  
output_tensor_shape : torch.Size([1, 32, 256, 256])
```